

Autodesk® Scaleform®

Scaleform

DirectX 9

Scaleform

3D

Ben Mowery

3.02

2011 4 14

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Autodesk® Scaleform® 4.2

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Scaleform 4.2

Scaleform Corporation
6305 Ivy Lane, Suite 310
Greenbelt, MD 20770, USA

www.scaleform.com

info@scaleform.com

(301) 446-3200

(301) 446-3199

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Scaleform
Adobe Flash Studio

Scaleform

Scaleform
Flash UI

Flash
Scaleform

3D

Scaleform

Flash

Scaleform

Scaleform

Scaleform 4.0

SDK ShadowVolume
renderer”

“cannot create

Scaleform Player

2

Scaleform

Scaleform

<https://gameware.autodesk.com/scaleform/developer/>

- Web Scaleform SDK
- PDF
-

Scaleform C++ API

- XML Scaleform XML
- Scale9 Grid Scale9Grid
- IME Scaleform IME Flash
- IME
- Scaleform

3

3.1

Windows Scaleform
,Scaleform installer DirectX Scaleform
SDK 4.2 C:\Program Files\Scaleform\GFx

3rdParty

Scaleform libjpeg zlib

Projects

Visual Studio

App\Samples

Apps\Tutorial

Bin

Flash

FxPlayer: Flash HUD
Samples: Flash FLA

Video Demo Scaleform
Win32: Scaleform Player
AMP: AMP Flash
GFxExport: GFxExport Flash

7

Include

Scaleform

Lib

Scaleform

Resources

CLIK

Doc

2

PDF

3.2

Scaleform

C:\Program Files

(x86)\Scaleform\GFx SDK 4.1\Projects\Win32\Msvc90\Demos\GFx 4.2 Demos.sln

Demos.sln

Visual Studio

.sln

D3D9_Debug_Static

GFxPlayer

GFx SDK 4.2\Bin\Win32\Msvc90\GFxPlayer\GFxPlayer_D3D9_Debug_Static.exe 文件的更新時間，確認已成功編譯並運行該程式。

Start → All Programs → Scaleform → GFx SDK 4.2 → Demo GFx

Player D3Dx

SWF

Scaleform

Flash

3.3 Scaleform Player SWF

Start → All Programs → Scaleform → GFx SDK 4.2 → GFx Players → GFx Player D3D9

C:\Program Files\Scaleform\GFx SDK 4.2\Bin\Data\AS2\Samples\SWFToTexture

3DWindow.swf

FSAA
 AA Scaleform EdgeAA

 DP

 EdgeAA

 Scaleform Player Flash CPU CPU
 Scaleform
 60 CPU CTRL+Y

 Scaleform debug release debug
 Scaleform

3.4

Scaleform 4.2 Scaleform\GFx SDK 4.2\Apps\Tutorial SLN
 Scaleform-> GFx SDK 4.2-> Tutorial Windows for Visual Studio 2005
 Visual Studio 2008 “ Debug ”



2

ShadowVolume

3.5 Scaleform

Scaleform	Visual Studio	\$(GFXSDK)	Tutorial SDK
	C:\Program Files\Scaleform\GFx SDK 4.2\		
\$(GFXSDK)			“ Msvc90 ”
Visual Studio 2008	Visual Studio 2010	Msvc90	Msvc10

Scaleform

\$(GFXSDK)\Src
\$(GFXSDK)\Include
Visual Studio “ Additional Include Directories ”

AS3 AS3
“GFx/AS3/AS3_Global.h” AS3
Scaleform LITE

\$(DXSDK_DIR)\Lib\x86
\$(GFXSDK)\3rdParty\expat-2.1.0\lib
\$(GFXSDK)\Lib\\$(PlatformName)\Msvc90\Debug_Static\
\$(GFXSDK)\3rdParty\zlib-1.2.7\Lib\\$(PlatformName)\Msvc90\Debug
\$(GFXSDK)\3rdParty\jpeg-8d\Lib\\$(PlatformName)\Msvc90\Debug

“ Additional Library Directories ”

"\$(DXSDK_DIR)\Lib\x86";
"\$(GFXSDK)\3rdParty\expat-2.1.0\lib";
"\$(GFXSDK)\Lib\\$(PlatformName)\Msvc90\debug";
"\$(GFXSDK)\3rdParty\zlib-1.2.7\Lib\\$(PlatformName)\Msvc90\Debug";
"\$(GFXSDK)\3rdParty\jpeg-8d\Lib\\$(PlatformName)\Msvc90\Debug"
Msvc90 Visual Studio

\$(DXSDK_DIR)\Lib\x86
\$(GFXSDK)\3rdParty\expat-2.1.0\lib
\$(GFXSDK)\Lib\\$(PlatformName)\Msvc90\Release\
\$(GFXSDK)\3rdParty\zlib-1.2.7\Lib\\$(PlatformName)\Msvc90\Release
\$(GFXSDK)\3rdParty\jpeg-8d\Lib\\$(PlatformName)\Msvc90\Release

“ Additional Library Directories ”

```
"$(DXSDK_DIR)\Lib\x86";
"$(GFXSDK)\3rdParty\expat-2.1.0\lib";
"$(GFXSDK)\Lib\$(PlatformName)\Msvc90\Release";
"$(GFXSDK)\3rdParty\zlib-1.2.7\Lib\$(PlatformName)\Msvc90\Release";
"$(GFXSDK)\3rdParty\jpeg-8d\Lib\$(PlatformName)\Msvc90\Release"
"$(GFXSDK)\3rdParty\libpng\Lib\$(PlatformName)\Msvc90\Release"
```

Msvc90

Visual Studio

Scaleform

```
libgfx.lib
libAS2.lib
libAS3.lib
libgfxexpat.lib
libjpeg.lib
zlib.lib
libpng.lib
libgfxrender_d3d9.lib
libgfxsound_fmod.lib
```

.vcproj

Scaleform include

Tutorial\Section3.5

3.5.1 AS3_Global.h and Obj\AS3_Obj_Global.xxx

```
:AS3_Global.h    Obj\AS3_Obj_global.xxx
AS3_Obj_Global.xxx    "    "ActionScript 3    swf
"    "(script)    ,    GlobalObjectCPP,    C++
    Scaleform VM

AS3_Global.h    ClassRegistrationTable
    AS3    C++    ,    ,
    ClassRegistrationTable,
AS3_Global.h

    ClassRegistrationTable    include
ClassRegistrationTable(    )    AS3_Global.h
```

, (,),
AS3 VM ,
“ ” ,

4

Scaleform 3D Unreal® Engine 3, Gamebryo™,
Bigworld®, Hero Engine™, Touchdown Jupiter Engine™, CryENGINE™, Trinigy Vision™
Scaleform

Scaleform DirectX DirectX ShadowVolume SDK
DirectX
DXUT 2D 3D

Scaleform Flash Scaleform DXUT

DXUT
Win32 DirectX Scaleform SDK GfxPlayerTiny.cpp

4.1 *Flash*

Flash 3D [Gfx::Loader](#)
Flash [Gfx::MovieDef](#) Flash [Gfx::Movie](#)
[Render::Renderer2D](#) [Render::HAL](#) DirectX
Scaleform API

ShadowVolume 4.1

#1:

ShadowVolume.cpp

```
#include "Gfx_Kernel.h"  
#include "Gfx.h"  
#include "Gfx_Renderer_D3D9.h"
```

Scaleform

GFxTutorial

Scaleform
Scaleform

```
//          GFx::Loader
Loader      gfxLoader;

//  SWF/GFx          GFxMovieDef
Ptr<MovieDef>  pUIMovieDef;

//          GFx::Movie
Ptr<Movie>  pUIMovie;

//  Renderer
Ptr<Render::D3D9::HAL>  pRenderHAL;
Ptr<Render::Renderer2D>  pRenderer;
MovieDisplayHandle      hMovieDisplay;
```

#2 GFx::System

Scaleform

[GFx::System](#)

Scaleform

WinMain

```
//          GFx::System
GFx::System      gfxInit;
```

GFx::System

Scaleform

Scaleform

WinMain

GFx::System

Scaleform

Tutorial

GFx::System

GFx::System

GFxTutorial

GFx::System

GFx::System

GFx::System::Init()

GFx::System::Destroy()

#3:

Scaleform

WinMain

InitApp()

InitApp()

```

gfx = new GFxTutorial();
assert(gfx != NULL);
if(!gfx->InitGFx())
    assert(0);

```

```

GFxTutorial      GFx::Loader      GFx::Loader
                SWF/GFx      SWF/GFx
                SWF/GFx      GFx::Loader
                GFx::Log

```

```

GFxTutorial::InitGFx()      GFx::Loader      GFx::Loader
                        SetLog      Scaleform
                        GFxPlayerLog
                        GFx::Log

```

```

//      -Scaleform
//
gfxLoader->SetLog(Ptr<Log>(*new PlayerLog()));

```

```

GFx::Loader      GFx::FileOpener
                GFx::FileOpener

```

```

//
Ptr<FileOpener> pfileOpener = *new FileOpener;
gfxLoader->SetFileOpener(pfileOpener);

```

```

Render::HAL      Scaleform      D3D9
                Scaleform      Scaleform
                InitHAL      D3D      Render HAL      IDirect3DDevice9
                Scaleform      DX9      UI

```

```

pRenderHAL = *new Render::D3D9::HAL();
if (!(pRenderHAL = *new Render::Render2D(pRenderHAL.GetPtr())))
    return false;

```

```

                Scaleform      Render::HAL      Render::D3D9::HAL
Scaleform

```

#4 Flash

```
GFx::Loader GFx::MovieDef
GFx::MovieDef

//
pUIMovieDef = *gfxLoader.CreateMovie(UIMOVIE_FILENAME,
                                     Loader::LoadKeepBindData |
                                     Loader::LoadWaitFrame1, 0);

LoadKeepBindData
D3D

LoadWaitFrame1 CreateMovie
GFx::ThreadTaskManager
```

[Memory System Overview](#)

#5

```
GFx::MovieDef GFx::Movie GFx::Movie

pUIMovie = *pUIMovieDef->CreateInstance(true, 0, NULL);
assert(pUIMovie.getPtr() != NULL);

CreateInstance false

Flash

Memory System

Overview

Advance() false CreateInstance

//
```

```
pUIMovie->Advance(0.0f, 0, true);
```

```
//
```

```
MovieLastTime = timeGetTime();
```

[Advance](#)

3D

```
pUIMovie->SetBackgroundAlpha(0.0f);
```

Flash

3D

#6

Scaleform

Render::D3D9::HAL

DirectX

D3D

Scaleform

Render::D3D9::HAL::InitHAL

/

D3D

InitHAL

ShadowVolume

OnResetDevice

DXUT

GFxTutorial

OnResetDevice

```
pRenderHAL->InitHAL(
```

```
Render::D3D9::HALInitParams(pd3dDevice, presentParams,
```

```
Render::D3D9::HALConfig_NoSceneCalls));
```

InitHAL()

D3D

Scaleform HAL_NoSceneCalls

Scaleform

DirectX

BeginScene()

EndScene()

ShadowVolume

OnFrameRender

#7

D3D

D3D

ShadowVolume OnLostDevice

Render::HAL

GFxTutorial

OnLostDevice

D3D

pRenderHAL->ShutdownHAL();

DXUT

Win32/DirectX

Scaleform SDK

GFxPlayerTiny.cpp

#8

Scaleform

GFxTutorial

WinMain

GFxTutorial

```
delete gfx;  
gfx = NULL;
```

DirectX 9

Scaleform

InitHAL()

ShutdownHAL()

DirectX 9

InitHAL

D3DPOOL_DEFAULT

InitHAL

D3DPOOL_DEFAULT

ShutdownHAL

D3DPOOL_DEFAULT

DXUT

ShadowVolume

DXUT

OnResetDevice

D3DPOOL_DEFAULT

OnLostDevice

GFxTutorial::OnLostDevice

ShutdownHAL

GFxTutorial::OnResetDevice

InitHAL

InitHAL

ShutdownHAL

D3DPOOL_DEFAULT

#9

D3D

到

GFxTutorial::OnResetDevice

```
//  
RECT windowRect = DXUTGetWindowClientRect();  
DWORD windowWidth = windowRect.right - windowRect.left;  
DWORD windowHeight = windowRect.bottom - windowRect.top;  
pUIMovie->SetViewport(windowWidth, windowHeight, 0, 0,
```

windowWidth, windowHeight);

[SetViewport](#)

PC

Scaleform

OpenGL

Scaleform

Flash

4.2

#10 DirectX

ShadowVolume OnFrameRender() D3D
BeginScene() EndScene() EndScene()
GfXTutorial::AdvanceAndRender()

```
void AdvanceAndRender(void)
{
    DWORD mtime = timeGetTime();
    float deltaTime = ((float)(mtime - MovieLastTime)) / 1000.0f;
    MovieLastTime = mtime;

    pUIMovie->Advance(deltaTime, 0);
    pRenderer->BeginFrame();

    if (hMovieDisplay.NextCapture(pRenderer->GetContextNotify()))
    {
        pRenderer->Display(hMovieDisplay);
    }

    pRenderer->EndFrame();
}
```

Advance() deltaTime
GfX::Movie::Advance

#11

[Render::Renderer2D::Display](#) DirectX D3D

D3D

Render::Renderer2D::Display D3D

Scaleform
DX9

DX9

GfxTutorial::AdvanceAndRender()

```
//    Gfx    DirectX
g_pStateBlock->Capture();

//
gfx->AdvanceAndRender();

//    DirectX
g_pStateBlock->Apply();
```

#12 **UI**

DXUT
ShadowVolume.cpp
DXUT

4.1

```
// Disable default UI
...
```

DirectX

Flash



3 GfX Flash ShadowVolume

4.2

GfX::Movie::SetViewport

Flash

Flash

Scaleform

-
-

4:3

Scaleform

800×600

1600×1200

Scaleform

Scaleform

[GfX::Movie::SetViewScaleMode](#)

GfXTutorial::InitGfX()

GfX::Movie

```
pUIMovie->SetViewScaleMode(Movie::SM_ShowAll);
```

SetViewScaleMode

SM_NoScale	Flash
SM_ShowAll	
SM_ExactFit	
SM_NoBorder	

[SetViewAlignment](#)

SM_NoScale SM_ShowAll

SetViewAlignment

```
pUIMovie->SetViewAlignment(Movie::Align_CenterRight);
```

SetViewScaleMode SetViewAlignment

SetViewAlignment

SetViewScaleMode

SM_NoScale

Scaleform

d3d9guide.fla

ActionScript

C++

◦ SetViewScaleMode

SetViewAlignment

ActionScript Stage

(State.scaleMode, Stage.align)◦

4.3

ShadowVolume

Scaleform

Flash

Flash

[GFx::Movie::HandleEvent](#)

GFx::Event

GFx::Movie

4.3.1

ShadowVolume

MsgProc

Win32

GFxTutorial::ProcessEvent

Scaleform

WM_MOUSEMOVE,

WM_LBUTTONDOWN

WM_LBUTTONUP

```
void ProcessEvent(HWND hWnd, unsigned uMsg, WPARAM wParam, LPARAM lParam,
                  bool *pbNoFurtherProcessing)
{
    int mx = LOWORD(lParam), my = HIWORD(lParam);
    if(pUIMovie)
```

```

{
    if(uMsg == WM_MOUSEMOVE)
    {
        MouseEvent mevent(GFx::Event::MouseMove, 0, mx, my);
        pUIMovie->HandleEvent(mevent);
    }
    else if(pMovieButton && uMsg == WM_LBUTTONDOWN)
    {
        ::SetCapture(hWnd);
        MouseEvent mevent(GFx::Event::MouseDown, 0, mx, my);
        pUIMovie->HandleEvent(mevent);
    }
    else if(pMovieButton && uMsg == WM_LBUTTONUP)
    {
        ::ReleaseCapture();
        MouseEvent mevent(GFx::Event::MouseUp, 0, mx, my);
        pUIMovie->HandleEvent(mevent);
    }
}
}

```

Scaleform

#1:

```

pMovie->SetViewport(screen_width, screen_height, 0, 0, screen_width,
                    screen_height, 0);

```

(0, 0)

Scaleform

#2

```

pMovie->SetViewport(screen_width, screen_height, 0, 0, screen_width / 4,
                    screen_height / 4, 0);

```

HandleEvent

Scaleform

#3

```

movie_width = screen_width / 6;
movie_height = screen_height / 6;

```

```
pMovie->SetViewport(screen_width, screen_height,
                    screen_width / 2 - movie_width / 2,
                    screen_height / 2 - movie_height / 2,
                    movie_width, movie_height);
```

```
(0,0) (screen_width /
2 - movie_width / 2, screen_height / 2 - movie_height / 2)
```

Flash [Gfx::Movie::SetViewAlignment](#)
[Gfx::Movie::SetViewport](#) SetViewAlignment [SetViewScaleMode](#)
Scaleform

4.3.2

[Gfx::Movie::HandleEvent](#)

[Gfx::KeyEvent](#)

[Gfx::CharEvent:](#)

```
KeyEvent(EventType eventType = None,
          Key::Code code = Key::None,
          UByte asciiCode = 0,
          UInt32 wcharCode = 0,
          UInt8 keyboardIndex = 0)
```

```
CharEvent(UInt32 wcharCode, UInt8 keyboardIndex = 0)
```

Gfx::KeyEvent	Gfx::CharEvent	ASCII	Windows
Gfx::CharEvent	WM_CHAR	Gfx::KeyEvents	WM_SYSKEYDOWN,
WM_SYSKEYUP, WM_KEYDOWN	WM_KEYUP		

- 'c' SHIFT : Gfx::KeyEvent WM_KEYDOWN
- 'c'
-
- SHIFT
- Gfx::CharEvent WM_CHAR “ ” ASCII 'C'
- Scaleform
- 'c' Gfx::KeyEvent WM_KEYUP
- Gfx::CharEvent

• F5 GfX::KeyEvent
 F5 ASCII GfX::CharEvent
 GfX::KeyEvent GfX_Event.h
 Flash GfXPlayerTiny.cpp Scaleform Player
 Windows Flash
 ProcessKeyEvent 3D

 void ProcessKeyEvent(Movie *pMovie, unsigned uMsg, WPARAM wParam, LPARAM lParam)

 Windows WndProc WM_CHAR, WM_SYSKEYDOWN, WM_SYSKEYUP,
 WM_KEYDOWN WM_KEYUP Scaleform pMovie

 GfX::KeyEvent GfX::CharEvent
 GfX::CharEvent ASCII Unicode
 IME . IME IME

 GfX::KeyEvent
 Page Up Page Down

 4.3
 3D "Settings" DX9 C++
 d3d9guide fla

 Scaleform Flash UI UI
 DXUT DXUT Scaleform
 Scaleform
 DXUT Scaleform Flash
 UI

 "Change Mesh"
 "Change Mesh" Flash

4.3.3

3D

UI 3D Gfx::Movie::HitTest 解决 Flash
中的 在处理完一个鼠标事件后，改变 GfxTutorial::ProcessEvent 以调用 [HitTest](#),

5

5.1

Scaleform

TTF

Flash

Windows

Linux

Scaleform

<http://gameware.autodesk.com/scaleform/developer/?action=doc>

5000

Scaleform

Scaleform

Scaleform

TTF

Scaleform

“

”

“

”

Bin\Data\AS2\Samples\FontConfig

sample.swf

Scaleform Player D3D9

早上好

///

你好!早上好!

///|///|

多想你呀!

///|

Chinese (繁體中文版)

3a

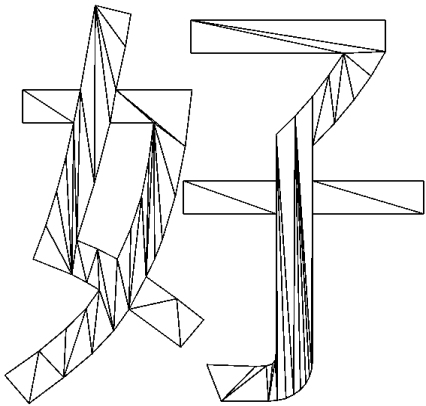
4b

/

CTRL+W



4c



4d

4c 4d Scaleform

Flash

gfx_2.1_texteffects_sample.zip

Isotropic shadow with offset

Text Shadow

文本陰影

Anisotropic glow, BlurX=2, BlurY=7

Text Glow 1

文本煥發 1

Isotropic glow, BlurX=BlurY=4, Knockout

TEXT GLOW 2

文本煥發 2

Rotating glow

Text Glow 3

文本煥發 3

Changing blurriness

Text Glow 4

文本煥發 4

6 C++ Flash

Flash

Flash

C++

Scaleform

Scaleform

C++

6.1

C++

Scaleform

C++

FSCommand

ExternalInterface

FSCommand

ExternalInterface

GfX::Loader

C++

FSCommand

fscommand

ExternalInterface

flash.external.ExternalInterface.call

[GfX::Value](#)

6.1.2

FSCommands

ExternalInterface

fscommands

gfxexport

SWF

fscommands

-fstree -fslist -fsparams

ExternalInterface

fscommands

6.1.1 FSCommand

fscommand

fscommand("setMode", "2");

fscommand

ExternalInterface

Scaleform FSCommand

[GfX::FSCommandHandler](#)

FSCommand

GfX::Loader

GfX::Movie

GfX::Movie

fscommand

GfXPlayerTiny

("FxPlayerFSCommandHandler")

ShadowVolume

6.1

GFx::FSCommandHandler

```
class OurFSCommandHandler : public FSCommandHandler
{
    public:
    virtual void Callback(Movie* pmovie,
                          const char* pcommand, const char* parg)
    {
        GFxPrintf("FSCommand: %s, Args: %s", pcommand, parg);
    }
};
```

Callback

fscommand

fscommand

GFxTutorial::InitGFx() GFx::Loader

```
//      FSCommand
Ptr<FSCommandHandler> pcommandHandler = *new OurFSCommandHandler;
gfxLoader->SetFSCommandHandler(pcommandHandler);
```

GFx::Loader

GFx::Movie

SetFSCommandHandler

fscommand

ShadowVolume

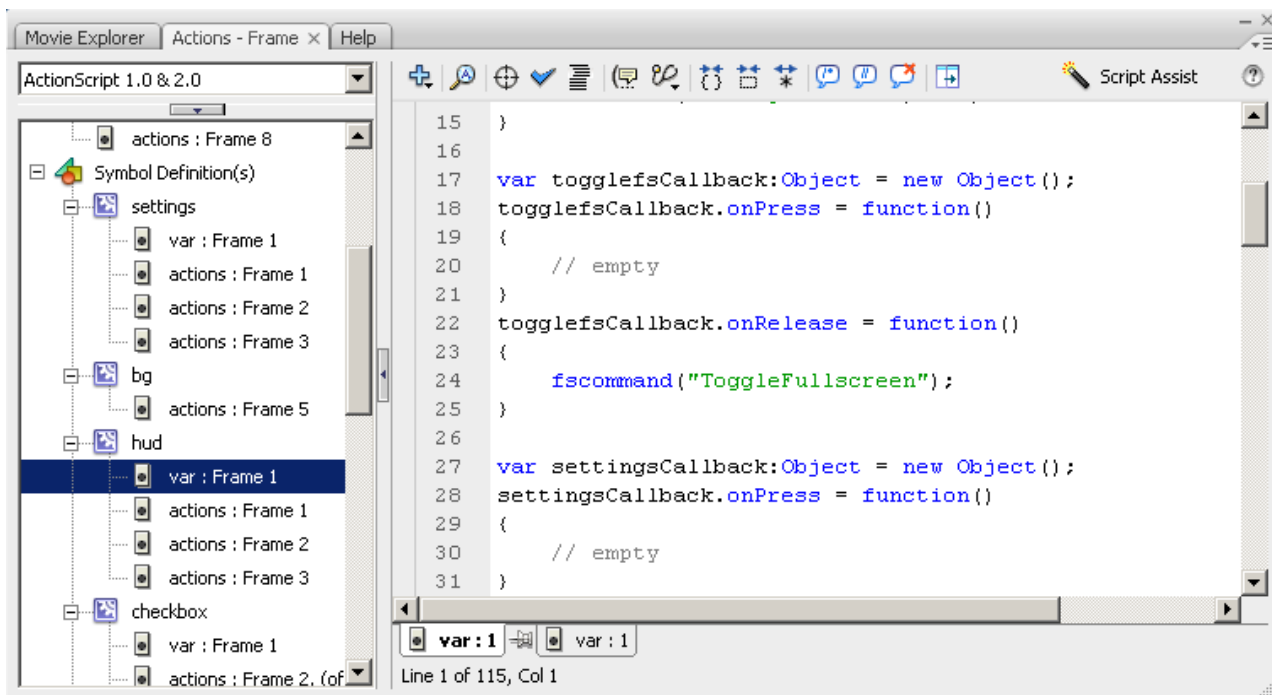
“Toggle Fullscreen”

FSCommand: ToggleFullscreen, Args:

Flash Studio d3d9guide.swf (F9)

ActionScript block for Symbol Definition(s) → hud → var : Frame 1

fscommand



6 fscommand()

	fscommand("ToggleFullscreen")	fscommand("ToggleFullscreen", this.UI._visible)
C++	this.UI._visible	flash (CTRL+ALT+Shift+S) d3d9guide.swf

FSCCommand

ShadowVolume

fscommand

```
if(strcmp(pcommand, "ToggleFullscreen") == 0)
    doToggleFullscreen = true;
```

DXUT OnFrameMove

OnFrameMove

```
if(doToggleFullscreen)
{
    doToggleFullscreen = false;
    DXUTToggleFullscreen();
}
```

Advance

Invoke

.

6.1.2 ExternalInterface

Flash ExternalInterface

fscommand

[ExternalInterface](#)

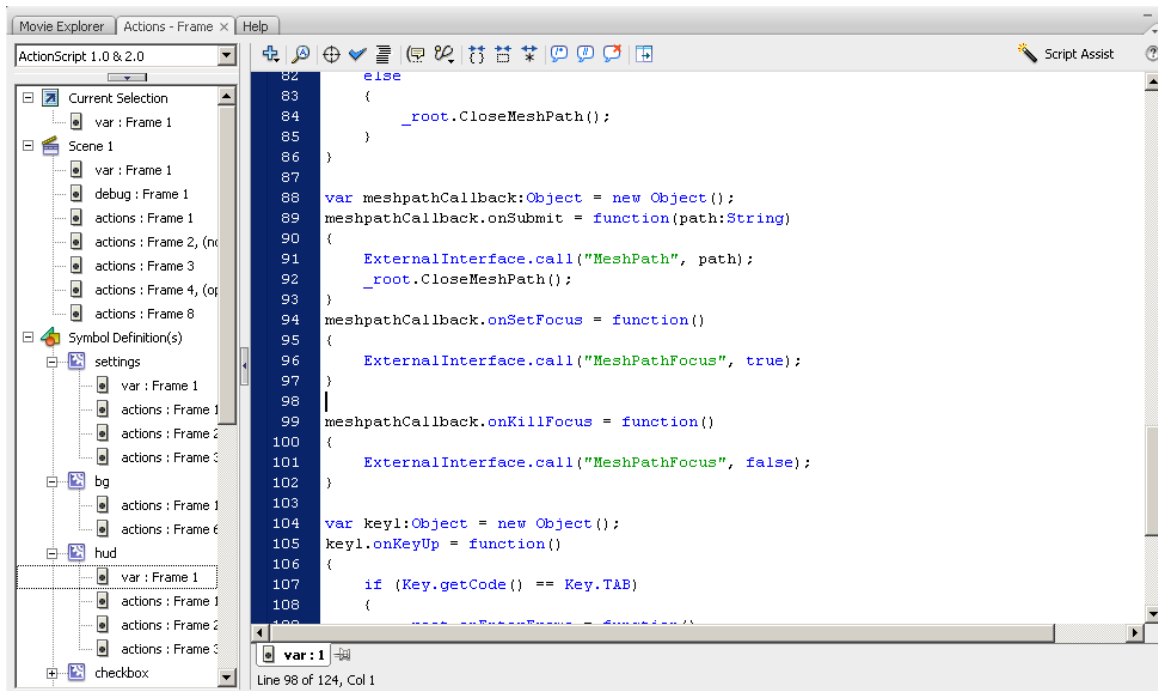
fscommand

```
class OurExternalInterfaceHandler : public ExternalInterface
{
public:
    virtual void Callback(Movie* pmovieView,
                          const char* methodName,
                          const Value* args,
                          unsigned argCount)
    {
        Printf("ExternalInterface: %s, %d args: ",
               methodName, argCount);
        for(unsigned i = 0; i < argCount; i++)
        {
            switch(args[i].GetType())
            {
            case Value::VT_Null:
                GFxPrintf("NULL");
                break;
            case Value::VT_Boolean:
                GFxPrintf("%s", args[i].GetBool()? "true" : "false");
                break;
            case Value::VT_Number:
                GFxPrintf("%3.3f", args[i].GetNumber());
                break;
            case Value::VT_String:
                GFxPrintf("%s", args[i].GetString());
                break;
            default:;
                GFxPrintf("unknown");
                break;
            }
            GFxPrintf("%s", (i == argCount - 1) ? "" : ", ");
        }
        GFxPrintf("\n");
    }
};
```

GFx::Loader

```
Ptr<ExternalInterface> pEIHandler = *new OurExternalInterfaceHandler;
gfxLoader.SetExternalInterface(pEIHandler);
```

ActionScript for Symbol Definitions() → hud → var : Frame 1



7 MeshPathFocus

(true false)

“MeshPathFocus”

```
ExternalInterface: MeshPathFocus, 1 args: false
Focus change:
    _level0.hud.text_MeshPath.field => null
ExternalInterface: MeshPathFocus, 1 args: true
Focus change:
    null => _level0.hud.text_MeshPath.field
```

GFxTutorial

```
f(strcmp(methodName, "MeshPathFocus") == 0 && argCount == 1)
{
    if(args[0].GetType() == Value::VT_Boolean)
        gfx->SetTextboxFocus(args[0].GetBool());
}
```

GFxTutorial::ProcessEvent

3D

```
if(uMsg == WM_SYSKEYDOWN || uMsg == WM_SYSKEYUP ||
    uMsg == WM_KEYDOWN    || uMsg == WM_KEYUP    ||
    uMsg == WM_CHAR)
{
    if(textboxHasFocus || wParam == 32 || wParam == 9)
    {
        ProcessKeyEvent(pUIMovie, uMsg, wParam, lParam);
        *pbNoFurtherProcessing = true;
    }
}
(ASCII 32) tab(ASCII 9) "Toggle UI" "Settings"
"Change Mesh"
```

OurExternalInterfaceHandler::Callback

```
static bool doChangeMesh = false;
static wchar_t changeMeshFilename[MAX_PATH] = L"";

...

if(strcmp(methodName, "MeshPath") == 0 && argCount == 1)
{
    doChangeMesh = true;
    const char *filename = args[0].GetString();
    MultiByteToWideChar(CP_ACP, MB_PRECOMPOSED, filename, -1,
        changeMeshFilename, _countof(changeMeshFilename));
}
```

DXUT OnFrameMove

DXUT

6.2 C++

6.1

C++

Scaleform

C++

Scaleform

C++

6.2.1

Scaleform [GetVariable](#) [SetVariable](#) 6.2
HUD (fxplayer.swf) Scaleform Player

F5

```
void GFxTutorial::ProcessEvent(HWND hWnd, unsigned uMsg, WPARAM wParam,
                               LPARAM lParam, bool *pbNoFurtherProcessing)
{
    int mx = LOWORD(lParam), my = HIWORD(lParam);

    if(pHUDMovie && uMsg == WM_KEYDOWN)
    {
        if(wParam == VK_F5)
        {
            int counter = (int)pHUDMovie->GetVariableDouble("_root.counter");
            counter++;
            pHUDMovie->SetVariable("_root.counter",
                                   Value((double)counter));

            char str[256];
            sprintf_s(str, "testing! counter = %d", counter);
            pHUDMovie->SetVariable("_root.MessageText.text", str);
        }
    }
    ...
}
```

[GetVariableDouble](#) _root.counter GetVariableDouble
[GFX::Movie](#) SetVariable _root.counter
 GetVariable SetVariable
fxplayer Flash MessageText
 HUDText _root.counter SetVariable

Flash TextField.text [GFX::Movie::Invoke](#)

6.2.2

Scaleform



8 SetVariable HUD

GFx::Value

SetVariable

GFx::Movie::SetVarType

“ ”

_root.mytextfield

3

SetVariable("_root.mytextfield.text", "testing",

SV_Normal)

1

SV_Sticky

3

_root.mytextfield.text

C++

SetVariableArray

Flash

GFxTutorial::InitGFx()

_root.SceneData

```
//
Value sceneData[3];
sceneData[0].SetString("Scene with shadow");
sceneData[1].SetString("Show shadow volume");
sceneData[2].SetString("Shadow volume complexity");
pUIMovie->SetVariableArraySize("_root.SceneData", 3);
pUIMovie->SetVariableArray(Movie::SA_Value,
    "_root.SceneData", 0, sceneData, 3);
```

ShadowVolume

6.2.2

UI

GfX::Movie::Invoke	WM_CHAR	openMeshPath
--------------------	---------	--------------

openMeshPath	1	1	Scaleform
--------------	---	---	-----------

initFirstFrame	true	1
----------------	------	---

[InvokeArgs](#) printf va_list Gfx::Value Invoke
 InvokeArgs printf vprintf

6.3 Flash

C++ SWF
 C++
 MMOG HUD HUD HUD
 HUD
 SWF C++
 Gfx::Movie
 SWF Gfx::Movie loadMovie unloadMovie
 C++ SWF Gfx::Movie
 MMOG _global.inventory SWF
 _global.inventory SWF unloadMovie

container.swf ActionScript container.swf
 MovieClipLoader ActionScripts

```
var mclLoader:MovieClipLoader = new MovieClipLoader();
```

ActionScript Flash

```
////  
//
```

```
function LoadFlashLevel(url, level)
{
    trace("LoadFlashLevel(" + url + ", " + level + ")\n");
    mclLoader.loadClip(url, level);
}
```

```

}

//
//

function LoadFlash(url, objectName)
{
    trace("LoadFlashLevel(" + url + ", " + objectName + ")\n");
    var container:MovieClip = createEmptyMovieClip(objectName,
                                                    getNextHighestDepth());
    mclLoader.loadClip(url, container);
}

```

“ level ”
Z UI

LoadMovieLevel Z
_levelN.variableName , level6.counter) 。

LoadMovie
 , root.inventoryWindow.widget1.counter) 。

Flash root root
level 6, root.counter level6.counter
LoadMovieLevel root
LoadMovie root.myMovie root.counter
 : lockroot = true。 Lockroot ActionScript
root root root
ActionScript Adobe Flash
GFX::Movie

Container fla

LoadMovie LoadMovieLevel loadClip

MovieClipLoader
ActionScript

Container.fla
ExternalInterface

C++

```

    .

//      :
//  1.
//  2.
//  3.
//  4.
//
//
//      LoadMovie()
//
//      .
//
//      ExternalInterface      C++
//      ActionScript

var mcListener:Object = new Object();

mcListener.onLoadError = function(target_mc:MovieClip, errorCode:String,
                                status:Number)
{
    trace("Error loading image: " + errorCode + " [" + status + "]");
};

mcListener.onLoadStart = function(target_mc:MovieClip) : Void
{
    trace("onLoadStart: " + target_mc);
};

mcListener.onLoadProgress = function(target_mc:MovieClip,
                                numBytesLoaded:Number,
                                numBytesTotal:Number) : Void
{
    var numPercentLoaded:Number = numBytesLoaded / numBytesTotal * 100;
    trace("onLoadProgress: " + target_mc + " is " +
        numPercentLoaded + "% loaded");
};

mcListener.onLoadComplete = function(target_mc:MovieClip,
                                status:Number) : Void
{
    trace("onLoadComplete: " + target_mc);
};
```


7

3D ,
 ,
 “ ” , Flash Studio SWF , SWF
 ,
 , UI
 UI



1. , cell.x , (cellwall.jpg)

UI ,
cellwallRT.jpg , MeshMaterialList
Material {
1.000000;1.000000;1.000000;1.000000;;
40.000000;

```

1.000000;1.000000;1.000000;;
0.000000;0.000000;0.000000;;
TextureFilename {
"cellwallRT.jpg";
}
2.          ,cell.x          DXUT          ,DXUT
          ,          CreateTexture
          ;          ,          (Render Texture)          ,
          DXSDK          ,
          ,          DXUT          ,
pd3dDevice->CreateTexture(rtw,rth,0,
D3DUSAGE_RENDERTARGET|D3DUSAGE_AUTOGENMIPMAP, D3DFMT_A8R8G8B8,
D3DPPOOL_DEFAULT, &g_pRenderTex, 0);

g_Background[0].m_pTextures[BACK_WALL] = g_pRenderTex;
ptex->Release();

3.          AdvanceAndRender          GFx          ,

a.          ,
pd3dDevice->GetRenderTarget(0, &poldSurface);
pd3dDevice->GetDepthStencilSurface(&poldDepthSurface);
b.          ,          ,
ptex->GetSurfaceLevel(0, &psurface);
HRESULThr = pd3dDevice->SetRenderTarget(0, psurface );
c.          ,          Display,          a

```

8 GFxExport

SWF SWF SWF SWF

GFxExport SWF

DDS DXT

GFxExport

Scaleform GFxExport

d3d9guide.swf fxplayer.swf Scaleform

```
gfxexport -i DDS -c d3d9guide.swf
gfxexport -i DDS -c fxplayer.swf
```

gfxexport.exe C:\Program Files\Scaleform\\$(GFXSDK)

convert.bat 7

-i DDS DDS DirectX DXT

DXT

-c Scaleform

DXT

-share_images

ShadowVolume 8 Scaleform

GFx::Loader::CreateMovie

9

Scaleform

- Flash 3 Scaleform Player SWF SDK Gamebryo
- [FAQs](#)
- Scale9 [Scale9Grid Overview](#)
- Scaleform SWF Flash [Gfx::FileOpener](#)
- [Gfx::ImageCreator](#)
- [Gfx::Translator](#)